

Topic 3.1

3.1: Comments on Reading Assignments for Module 2

This is a somewhat long lesson that illustrates many essentials of classes. The *class* is the mechanism we use to associate the operations that manipulate an object with the data that represents the state of the object. Operations that manipulate the state of the object, which is recorded in *data members*, are called *member functions*.

Note that constructors do not have return values. This is because they are constructing an object from the raw materials (memory). Constructors are invoked automatically and help us avoid uninitialized variables.

Public members of classes are those data members and member functions that are accessible by any other component of the program. The public members embody the logical view of the class as seen by client software (other parts of the program that use the class). Generally, we want to minimize the number of public data members. This facilitates swapping in alternate physical implementations for efficiency, debugging, etc.

Private members are those data members and member functions over which we want to maintain tight control: We only want to allow privileged parts of the program to access them. The privileged parts turn out to be exactly the member functions of the class itself (with a couple of extensions we'll learn about later).

Topic 3.2 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-3-dot-2>)